IF……Else

1. Write a c program to demonstrate getchar() and putchar().
2. Write a c program to demonstrate gets() and puts().
3. Write a c program to demonstrate getche() and putch().
4. Write a program to find out maximum number from two variables.(using simple if…..else)
5. Write a c program to find out minimum from two numbers.(using simple if…..else)
6. Write a c program to check whether a person is eligible for vote or not.(using simple if…..else)
7. Write a c program to check whether the number is odd or even.(using simple if…..else)
8. Write a c program to find out maximum from three numbers.(using simple if…..else)
9. Write a program to check whether the year is leap year or not.
10. Write a c program to check whether the number is negative or positive.
11. Write a c program to find out minimum from three numbers.(using simple if…..else if……else)
12. Write a c program to find out middle value from three numbers.(using simple if…..else if……else)
13. Write a program to check whether the student is pass or fail.(using simple if…..else if……else)
14. Write a c program to display average of a student when student is pass.(using simple if……else)
15. Write a c program to check whether the number is divisible by 3 or not.
16. Write a c program to check whether the number is divisible by 3 but not divisible by 9.
17. Write a c program to demonstrate conditional operator.
18. **Write a c program to prepare a mark sheet of a student for consider following criteria. (using else if ladder)**
    a. Give a class when student got >=40 marks in each subject.
    b. Pass class : per>=40 and <=48.
    c. Second class: per>=49 and <=59.
    d. First class : per>=60 and <=69.
    e. Distinction : per>=70.

19. Write a menu based program for the following operations.(Using if…….else ladder).
    1. Addition.
    2. Subtraction.
    3. Multiplication.
    4. Division.
    5. Remainder.

20. **Write a c program for sum of two numbers without using '+' sign.**
21. **Write a c program for subtraction of two numbers without using '-' sign.**
22. **Write a c program to find out remainder without using '%' sign.**
23. **Write a c program to find out maximum number without using '>' sign.**
24. **Write a c program to find out minimum number without using '<' sign.**
25. **Write a c program to find out middle value without using if …..condition (using ternary operator).**

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Switch Case

1. .write a menu based program for the following operations.(Using switch case)
    1. Addition.
    2. Subtraction.
    3. Multiplication.
    4. Division.
    5. Remainder.

2. Write a program to demonstrate &,~,^,<<,>> operator using switch case.

1. Write a c program to find out addition, multiplication, division, substraction, and remainder using following symbol which inputted by a user.(+,*,/,-,%).
2. Write a c program to display following message when user input following number.
    a. 1 Monday
    b. 2 Tuesday
    c. 3 Wednesday
    d. 4 Thursday
    e. 5 Friday
    f. 6 Saturday
    g. 7 Sunday
    h. default: invalid input.
3. Write a c program for signal as per follow.
    - If user input ‘R’ or ‘r’ then display: “stop traffic is jam”.
    - If user input ‘G’ or ‘g’ then display : “Go ahead ”
    - Default : “not permitted”.

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Loop

1. Write a c program to display your name 5 times(using Entry control loop).
2. Write a c program to display your name 10 times(using Exit Control Loop).
3. Write a c program to make a table of any numbers(using any loop).
4. Write a c program to find out Compound interest.
5. Write a c program to find out ascii value of A-Z.
6. Write a c program to find out ascii value of a-z.
7. Write a c program to find out ascii value of 0-9.
8. Write a c program for sum of 10 numbers.(using while loop).
9. **Write a c program for sum of n numbers.(Using do……while loop).**
10. Write a c program for sum of a given number.(using for loop).
11. **Write a c program for sum of odd numbers and multiplication of even numbers.(Using any loop).**
    i. Eg.12345
    ii. Sum of add=9.
    iii. Multiplication of even:8.
12. **Write a c program to find out factorial of given number.**
13. To read any five numbers and print the average value (while loop).
14. Write a program to find out all the numbers divisible by 5 and 7 between 1 to 100 (Do…while).
15. **Write a program to find out sum of given number (for loop).**
    a. Input : 6
    b. Output :21
16. **To find out total number of odd digits within the given numbers and print the sum of all odd digits (goto).**
    i. Ex. Input: 123456
    ii. Output:
        1. Total odd numbers: 3.
        2. Sum of odd number: 9.
17. Write a program for reverse a number using all loops (using switch case).
18. Write a program to find out sum of first and last digit of a given number.
    a. Input : 123456
    b. Output :7
19. **Write a program to find out largest digit from the given integer number.**
    a. Input :19863

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b. Maximum:9

20. Write a program to find out minimum digit from the given number.
    a. Input :12953
    b. Minimum :1
21. Write a program to find out average of three numbers (goto).
22. Read five persons height and weight and count the number of persons having height greater than 170 and weight less than 50.
23. **Check whether the given number is perfect or not. a number is perfect if its sum of digits is same as multiplication of digits.**
    a. Input : 123 1+2+3=6 1*2*3=6 perfect.
    b. Input: 234 2+3+4=9 2*3*4=24 not perfect.
24. **Check for Armstrong number. A number is Armstrong if sum of cube of every digit is same as the original number.**
    a. Input: $153=1^3+5^3+3^3=153$. Armstrong.
    b. Input: $356=3^3+5^3+6^3=368$.
25. Write a program to check whether the given number is prime or not.
26. **Write a program to find out and print all prime numbers lying between 50 to 150.**
27. **Twin primes are defined to be two consecutive odd numbers which are both primes. For example 11 and 13 are twin primes. Write a program to generate all the twin primes in a given range.**
28. Write a program to check whether the given number is binary or not.
29. Write a program to check whether the given number is octal or not.
30. Write a program to print all integer number from n1 to n2 (n1<n2) which are divisible by 3 but not divisible by 6. Count such numbers. Print six numbers in one output line. Also print count.
31. **Write a program for magic number. Sum of all digits (until the number becomes single digit) should be 1**.
    a. Input: 101 not magic.
    b. Input: 640 magic.

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Series

1) 1+2+3+4+5+…….+n.
2) 1+3+5+7+9+……..+n.
3) 2+4+6+8+……..+n.
4) 1-2+3-4+5-6+……n.
5) $1^2 +2^2+3^2+4^2+5^2$……+n^2.
6) 1-4+9-16+25-36……..+n^2. (Find out sum and display this series).
7) 1!+2!+3!+4!.....n!.
8) 1/1!+1/2!+1/3!+1/4!.....1/n!.
9) $X+x^2+x^3+x^4$+……x^n.
10) 1+2+4+8+16+32…….n.
11) 1+1/4+1/9+1/16…….n.
12) $1/1^2-1/2^2+1/3^2-1/4^2$+……..n.
13) 1+8+27+64+……n.
14) W.a.p to display fibonancis series up to n.(0 1 1 2 3 5 8 13 21….n).

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Pattern

1. * ** *** **** *****	*2.* * ** *** **** *****	*3.* ***** **** *** ** *	*4.* ***** **** *** ** *
5. 1 12 123 1234 12345	*6.* 1 12 123 1234 12345	*7.* $$$$$ $$$$ $$$ $$ $	*8.* $$$$$ $$$$ $$$ $$ $
9. **12345** **1234** **123** **12** **1**	*10.* **54321** **4321** **321** **21** **1**	*11.* **1** **21** **321** **4321** **54321**	*12.* **1** **21** **321** **4321** **54321**
13. 54321 4321 321 21 1	*14.* 12345 1234 123 12 1	*15.* 54321 5432 543 54 5	*16.* 12345 2345 345 45 5
17. 5 45 345 2345 12345	*18.* 5 54 543 5432 54321	*19.* 55555 4444 333 22 1	*20.* 55555 4444 333 22 1
21. 11111 2222 333 22 1	*22.* 5 44 333 2222 11111	*23.* 5 44 333 2222 11111	*24.* 1 22 333 4444 55555
25. 1 22 333 4444 55555	*26.* 1 1 4 1 4 9 1 4 9 16 1 4 9 16 25	*27.* 1 1 4 1 4 9 1 4 9 16 1 4 9 16 25	*28.* **1** **2 2** **3 3 3** **4 4 4 4** **5 5 5 5 5**
29. 5 4 4 3 3 3 2 2 2 2 1 1 1 1 1	*30.* a a b a b c a b c d a b c d e	*31.* * * * * * * * * * * * * * * *	*32.* **a b c d e** **a b c d** **a b c** **a b** **c**
33. $ $ $ $ $ $ $ $ $ $ $ $ $ $ $	*34.* AAAAA BBBB CCC DD E	*35.* AAAAA BBBB CCC DD E	*36.* E DD CCC BBBB AAAAA

37. ABCDE ABCD ABC AB A	*38.* A AB ABC ABCD ABCDE	*39.* A BA CBA DCBA EDCBA	*40.* EDCBA EDCB EDC ED E

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41.
```
ABCDE
 BCDE
  CDE
   DE
    E
```

42.
```
E
ED
EDC
EDCB
EDCBA
```

43.
```
A
A B
A B C
A B C D
A B C D E
```

44.
```
A
A C E
A C E G I
A C E G I K M
A C E G I K M O Q
A C E G I K M
A C E G I
A C E
A
```

45.
```
Z
Z Y
Z Y X
Z Y X W
Z Y X W V
```

46.
```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

47.
```
$$$$$$$$$$$
$         $
$         $
$         $
$         $
$         $
$$$$$$$$$$$
```

48.
```
0
101
21012
3210123
432101234
54321012345
```

49.
```
0
-1 0 1
-2 -1 0 1 2
-3 -2 -1 0 1 2 3
-4 -3 -2 -1 0 1 2 3 4
-5 -4 -3 -2 -1 0 1 2 3 4 5
```

50.
```
*
* *
*   *
*     *
*       *
*     *
*   *
* *
*
```

51.
```
*********
****  ****
***    ***
**      **
*        *
```

52.
```
*********
****  ****
***    ***
**      **
*        *
**      **
***    ***
****  ****
*********
```

53.
```
1
01
101
0101
10101
```

54.
```
ABCDEFGFEDCBA
ABCDEF   FEDCBA
ABCDE     EDCBA
ABCD       DCBA
ABC         CBA
AB           BA
A             A
```

55.
```
*
*1
*1*
*1*2
*1*2*
*1*2*3
```

56.
```
1
**
22
***
333
****
4444
*****
```

57.
```
1        1
12      21
123    321
1234  4321
123454321
```

58.
```
1
2  3
4  5  6
7  8  9  10
11 12 13 14
```

59.
```
      1
    2 1 2
  3   1   3
4     1     4
  3   1   3
    2 1 2
      1
```

60.
```
      1
    1   1
  1   2   1
1   3   3   1
1   4   6   4   1
1 5  10  10 5  1
```

61.
```
*
1*1
*1*1*
1*1*1*1
*1*1*1*1*
```

62.
```
1
*1*
3*1*3
*3*1*3*
5*3*1*3*5
```

63.
```
*
1 * 1
* 1 * 1 *
1 * 1
*
```

64.
```
5 * 3 * 1 * 3 * 5
4 * 2 * 2 * 4
3 * 1 * 3
2 * 2
3 * 1 * 3
4 *2 *2 * 4
5 * 3 * 1 * 3 * 5
```

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65.

```
4 * 2 * 2 * 4 *
 * 3 * 1 * 3 *
  * 2 * 2 *
   * 1 *
    *
   * 1 *
  * 2 * 2 *
 * 3 * 1 * 3 *
* 4 * 2 * 2 * 4 *
```

66.

Create a swastika using if condition.

```
*      ********
*      *
*      *
*      *
**************
       *      *
       *      *
       *      *
* * ****      *
```

Pointer

1. W.a.p to display address of three variables.
2. W.a.p. to display value of variable using pointer (any three variables).
3. W.a.p to find out sum of two integers without using (+) operator.
4. W.a.p to find out subtraction of two value without using (-) operator.
5. W.a.p to find out multiplication of two values without using (*) operator.
6. W.a.p to find out remainder of two values without using (%) operator.
7. W.a.p to find out max number without using (>,<) operator.
8. W.a.p to find out min number without using (<,> and and any loop).
9. W.a.p to find out middle value of three variables without using(if else statement).
10. W.a.p to swap two values without using third variables.
11. W.a.p to find out sum of given no(ex.12345 sum=15.)
12. W.a.p to find out factorials of given no.
13. W.a.p to display address of characters.
14. W.a.p to display character by character using pointer.
15. W.a.p to insert five character and sort in descending order.
16. W.a.p to insert five names and sort in ascending order.
17. W.a.p to insert 3X3 matrix and display its address and value.
18. W.a.p to find out sum of 3X3 matrix.
19. W.a.p to find out multiplications of 3X3 matrix.
20. W.a.p to demonstrate use of pointer to pointer(chain pointer).
21. W.a.p to demonstrate use of void pointer.
22. W.a.p to demonstrate use of call by reference.

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Function

1. W.a.p to find out sum of two values (using all types of function).
2. W.a.p to find out factorials of given number.
3. Write a function to find out average of five number
4. Write a function to check whether the given no is Armstrong or not.
5. Write a function to check whether the given no is magic or not.
6. Write a function to check whether the given no is prime or not.
7. Write a function to display prime no between 200-700 , ten prime each per line.
8. W.a.function for menu based program for sub, add, mul, div,mod.
9. W.a.p to find out sum of even and mul of odd no (eg. 12345 sum=6 mul=15).
10. W.a.p to find out uppercase, lowercase, digits, spaces, all characters.(menu based).
11. Write a program to find out factors of given number.
12. Write a program to sort out array in ascending and descending order.
13. Write a program to swap to values using call by reference.
14. Write a function for decimal to binary.
15. Write a function for binary to decimal.
16. Write a function for binary to octal.
17. Write a function for octal to decimal.
18. Write a function for decimal to octal.
19. Write a function for decimal to hexadecimal.
20. Write a function for binary to hexadecimal.
21. Write a function to find out factorials of no using recursion.
22. Write a function to display Fibonacci series using recursion.
23. Write a function to display following series and display sum using recursion.
    a. 1+2+3+4+5+….n sum=15.
    b. 1-2+3-4+5……n sum=3.
24. Write a program for menu based as per following information.
    a. Insert a string.
    b. Find out reverse.
    c. Check out palindrome or not.
    d. If palindrome then count numbers of character.
    e. If not palindrome than sort into descending order.
25. Write a function to sort out 10 names in ascending order.
26. Write a menu based program as per following information.
    a. Create function for initialize two matrices 3X3.
    b. Create a function for sum of matrix.
    c. Create a function for mul of matrix.
    d. Create a function for display sum of matrix.
    e. Create a function for display mul of matrix.

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File Management

1. Write a program to input your name and display or insert in name.txt file.
2. Write a program to read string from name.txt and display on terminals.
3. Write a program to insert a data in first.txt and display data on second.txt plus one character

   For example (bca input in first.txt) (cdb stored in second.txt).
4. Write a program to find out uppercase and lowercase letters and stored in uppercase.txt and lower case.txt respectively.
5. Write a program to input multiple lines in a file name all.txt and convert all characters into uppercase, lowercase and toggle case and stored in upper.txt, lower.txt and toggle.txt respectively.
6. Write a program to read a particular file(suppose file is exist) and open in append mode and display data which already exist and on other data which you edit.

Question Bank :-

Descriptive Question :

Unit :-1

1) Difference between High Level and Low Level language.
2) Difference between Compiler and Interpreter.
3) Why and when we need #include directive?
4) What is program?
5) Define compiler.
6) What is Header file?
7) When “statement missing” error occurs?
8) Explain printf (), scanf (), getch (), main ().
9) Difference between Syntax error and Runtime error.
10) What is meaning of #include<stdio.h> and #include<conio.h> statement?
11) Give the name of High Level Languages.
12) What is computer language ? Generation or types of computer Language.
13) Explain History of C Language.
14) Explain Structure of C program.
15) What is compiler? Explain compiler, interpreter, translator and assembler.
16) What is Character Set? Explain in details.
17) What is Tokens? Explain in details.
18) What is Keywords and Identifier? Explain with example.
19) What is constant? Explain it’s Types in Details.
20) What is Variable? Discuss its Rule in Details.
21) What is Data Type? Explain User Defined Data type in details.
22) What are the Basic Data Type available in C ? Explain in brief.
23) What is Data Type? Explain Derived Data type with example.
24) Draw a Flowchart for creation, compilation, and Execution of C program.

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25) Explain storage class specification.
26) Write down the steps to execute a C program.
27) What is Flowchart?
28) What is Algorithm?
29) Explain each symbols of flowchart.
30) Describe the rules for developing flowchart.
31) What are the rules for developing flowchart?
32) Differentiate flowchart and algorithm.
33) What are the difference between flowchart and algorithm?
34) Explain Flowchart with example.
35) Explain Algorithm with example.
36) Draw a flowchart for calculate simple interest
37) Draw a flowchart for find out Basic Salary.
38) Draw a flowchart to find out the person is eligible for voting or not.
39) Draw a flowchart to find out maximum number from three numbers
40) Draw a flowchart to print the sum of first 10 natural numbers.
41) Draw a flowchart to display the sum of digits for a given number.
42) Draw a flowchart for sum=1+3+5+……..n. terms.
43) Write an algorithm for calculate simple interest
44) Write an algorithm for find out Basic Salary.
45) Write an algorithm to find out the person is eligible for voting or not.
46) Write an algorithm to find out maximum number from three numbers
47) Write an algorithm to print the sum of first 10 natural numbers.
48) Write an algorithm to display the sum of digits for a given number.
49) Write an algorithm for sum=1+3+5+……..n. terms.
50) What is Trigraph character Explain it.
51) What are keywords? List out keywords available in C.
52) What is variable? Give the rules of defining identifiers.
53) Define C Tokens? Explain each with example.
54) What is Data Overflow and Data Underflow? Explain With Example.
55) Give the purpose of %i, %e, %o, %x format specifies.
56) What is type conversion? Explain implicit and explicit type conversion.

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Unit :-2

1) What is operator? list out operators available in c.
2) Explain Arithmetic Operators with example.
3) Explain Relational operators.
4) List out logical operators and explain in details.
5) What is assignment operator? Explain in brief.
6) What is increment and decrement operator?
7) Differentiate: increment and decrement operators.

8) What is conditional operator? or What is Ternary operator ? or what is the meaning of ?: operator ?
9) What is bitwise operator? Explain with example.
10) What are the special operators are available in c ? Explain any two.
11) Explain: Associatively or Priority table of Operator.
12) Explain following functions with examples?
    a) getch() b) tolower() c) toupper() d) isupper()
    e) getchar() f) gets() g) putchar() h) puts() i) toascii()
13) What is Decision Making? Explain simple If statement with example.
14) Explain Nesting of IF….Else Statement with example.
15) Explain Else If Ladder with example.
16) Explain Switch Statement with example.
17) Give the rules of switch case statement.
18) Explain Branching statement with example.
19) Difference between If….Else and Switch Case.
20) Explain Types of loop with Example.
21) Explain pretest and posttest?
22) Give the Difference between Entry and Exit Control Loop.
23) "Programmer should avoid the use of goto statement?" Why?
24) Why continue and break keywords are used in C ? Explain with example.

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Unit : 3

1) Explain difference operations performed on 1-D array.
2) Explain Two Dimensional Array or 2-D Array.
3) Explain compile time & runtime Initialization of 1-D Array.
4) Can we initialize 2-D Array in Compile time and runtime? How?
5) Explain Different Operations performed on 2-D Array.
6) Explain multi-dimensional & dynamic Arrays.
7) What is Array? Explain one dimensional Array with example.
8) What is Array? Explain two dimensional Arrays with example.
9) What do you mean by Array? Explain declaration and compile time initialization of one dimensional and two dimensional arrays.
10) Write a detail note on Array and its types.
11) What is bound array and unbounded array.

Unit :- 4

1) What is character Array? How we can declare & initialize character Array?
2) Why we put Null character at the end of String ?
3) How we can accept string from Terminal or Monitor.
4) How we can write a String on Terminal with Example.

5) Explain string Related functions available in C with example.
6) How we can create String Array?
7) Explain memory layout in array.
8) Explain the various categories of functions.
9) Which are the functions available in c to get a string from terminal?
10) Difference between %s and edit set conversion code?
11) What is substring? Explain substring function available in c?

Unit :-5

1) Explain Structure with example.
2) What is structure? Explain defining, declaring, & initializing structure in details.
3) What is difference between array and structure?
4) How we can initialize array of structure and structure of array?
5) Differentiate: structure and Union.
6) Difference between Structure and Array.
7) What is union? Explain with example.

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